University of Cape Town
Department of Computer Science

Computer Science CSC3003S
Class Test 1, 2019

Answer all questions.
Approximate marks per question are shown in brackets.

Time: 40 minutes **Marks: 40**

Section 1. Project Management [23 Marks]

Question 1 [9 Marks]

a. What is meant by *free float* in Critical Path Analysis? [1]

ANS: The amount of time an activity can expand [1/2] without affecting the start/finish of any other activity [1/2]

b. What is test-driven development? How does it differ from more traditional software development methods? [5]

ANS: Test Driven Development (TDD) = test first development + refactoring [1]. So TDD turns traditional development around and instead of writing functional code first and then your testing code as an afterthought [1], you first write your test code before your functional code [1]. Development occurs in very small steps: one test and a small bit of code at a time [1]. Once the code passes the test (i.e. it works), it is

refactored to ensure that it's remains of high quality [1]. After refactoring the tests are applied again before moving on [1]. The radical point= new code is only written when an automated test fails [1]

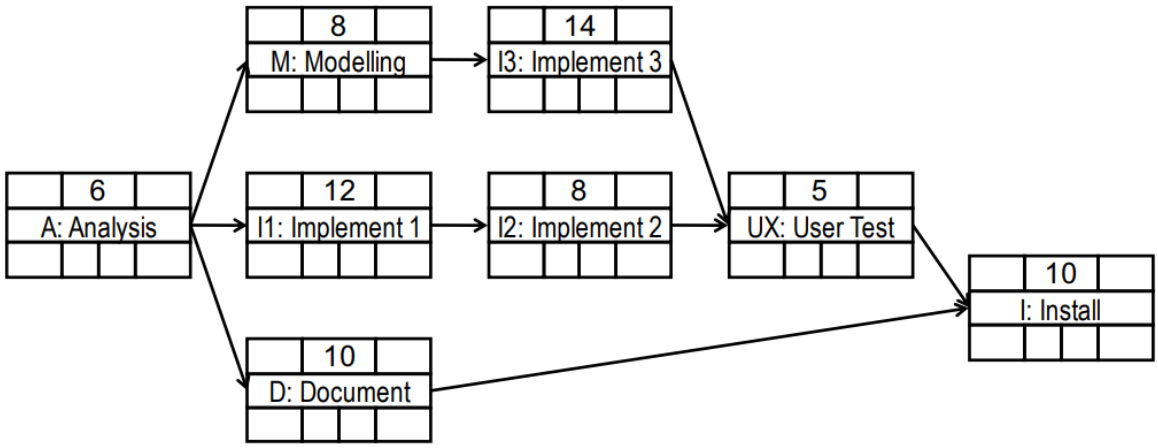
c. List 3 benefits of pair programming: [3]

ANS: Three of :

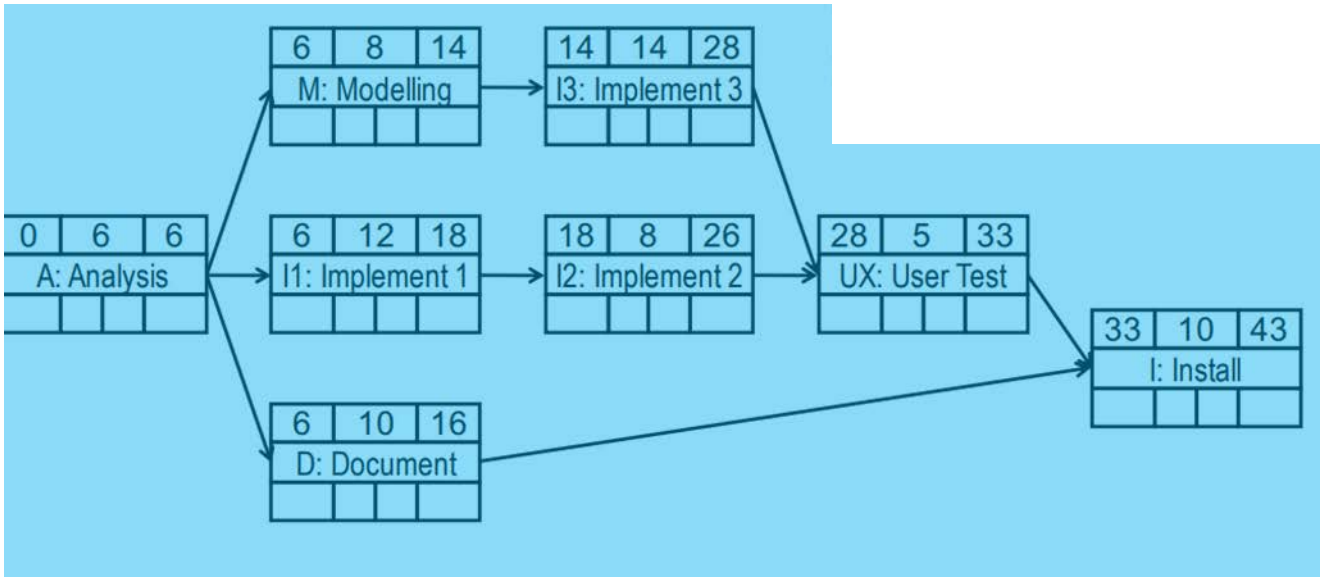
- Common ownership of code [1]
- Collective responsibility for the system[1]
- Spreads knowledge across the team. [1]
- Motivates refactoring [1]

Question 2 [14 Marks]

a. Fill in the 'forward pass' for the network analysis shown below: [8]

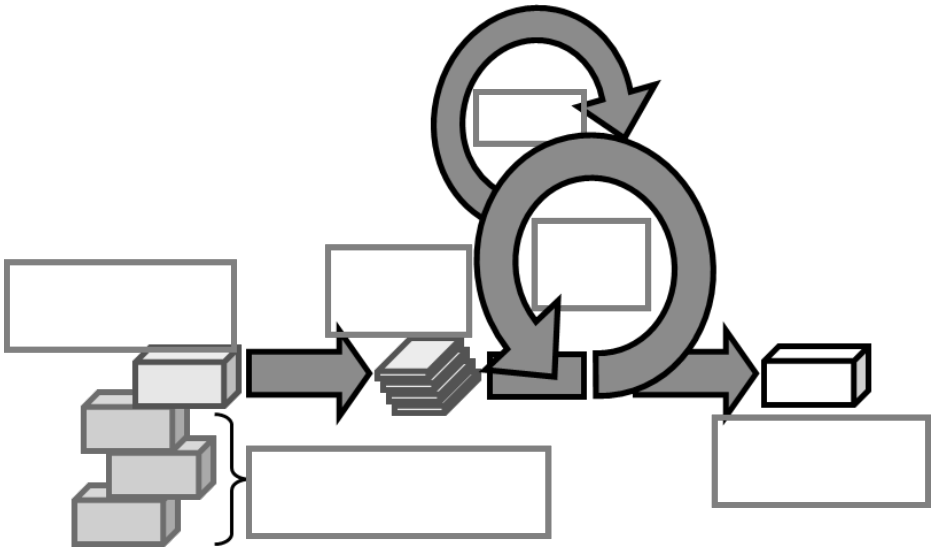


ANS:

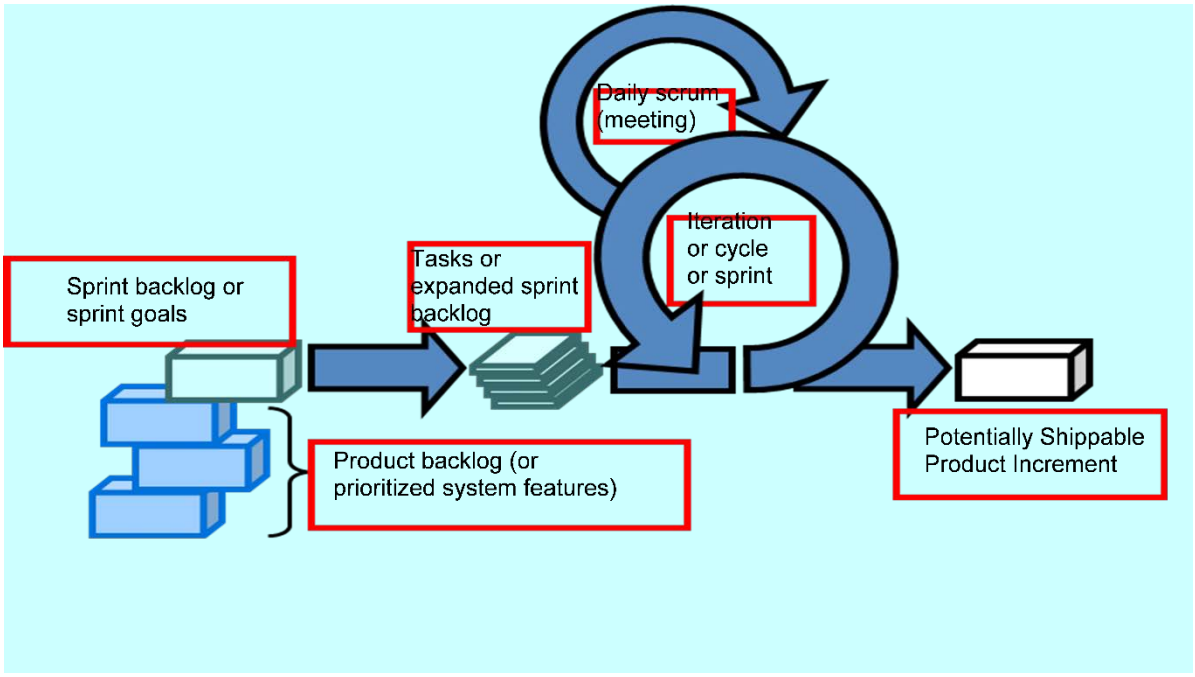


[1] for each correct top row

b. Provide the missing labels for the following diagram. [6]



ANS: 1 mark for each correctly labelled cell below



Section 2. Architecture [17 Marks]

Question 3 [5 Marks]

a. Cohesion and coupling are commonly referred to when talking about software architectural design. Explain what each of those two concepts mean, and what the goal of architectural design is in terms of those concepts. [3]

ANS:Cohesion is the degree to which communication takes place within the module [1]

Coupling is the degree to which communication takes place between modules [1]

Architectural design goal is about minimizing coupling while maximizing cohesion [1]

b. An online e-commerce system is an example of which kind of general application architecture? [1]

ANS: Transaction processing system [1]

c. Briefly explain the Information Expert design pattern. [1]

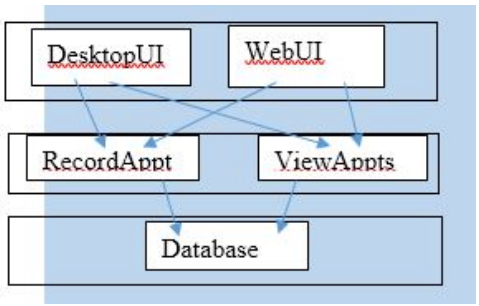
ANS: Answer: the class that has the information necessary, fulfils the responsibility. OR

"That which has the information, does the work." [1]

Question 4 [8 Marks]

You've recently been appointed as the software architect of a software system for a new start up which produces a time management application which has desktop interface as well as a web-based interface. The application should have functionality which records appointments and also views upcoming appointments. All appointments are stored in a database. Briefly explain what the most sensible underlying architectural design pattern to use is, why you chose it, and use a UML diagram to illustrate the actual system showing all the major components/modules. [8]

ANS:
Answer: The most sensible architecture to use would be a layered architecture [1], since there are different user interfaces with common functionality all linking to the same database. [1]



2 for components/modules, 2 for layers, 2 for dependencies/coupling

Question 5 [4 Marks]

a. What are the typical challenges of developing systems in the area of Information and Communication Technologies for Development (ICT4D)? [3]

ANS:
• Limited connectivity
• Variable literacy of users
• Stakeholders with competing interests
• Scale with limited resources
• Sustainability
// **** any three of these ****

b. What mechanism is used in Progressive Web App development, which acts as a proxy between the network and the application, allowing the caching of data locally so that the user can work offline. [1]

ANS: Service Workers [1]

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